AnsyBeam User Manual

MATLAB-based Euler–Bernoulli Beam Analysis Tool

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1 Introduction

AnsyBeam is a MATLAB-based educational tool for planar beam bending analysis using one-dimensional Euler–Bernoulli beam elements. Through editable input tables, users can define the beam geometry, material and sectional properties, support conditions, nodal loads, nodal moments, and uniformly distributed loads. The app then calculates nodal deflections, rotations, support reactions, and element end forces, and provides visual outputs showing the verified beam model, deflected shape, shear force diagram, and bending moment diagram.

2 Coordinate System and Sign Convention

AnsyBeam uses a one-dimensional beam axis aligned with the global x-direction. Each node is defined by its horizontal coordinate x. The transverse displacement is denoted by v, and the nodal rotation is denoted by θ.

The following sign convention is used:

- Positive vertical force is upward.
- Positive vertical displacement v is upward.
- Downward point loads and uniformly distributed loads (UDLs) are entered as negative values.
- Positive nodal rotation θ follows the mathematical counter-clockwise convention.
- Positive applied nodal moment is counter-clockwise.
- Clockwise applied nodal moment is entered as a negative value.
- Positive shear force follows the convention that the right-hand side of a cut section acts downward; equivalently, upward shear on the right-hand side of a cut section is negative (Right – Up – Negative: RUN).
- Positive bending moment corresponds to sagging bending, while negative bending moment corresponds to hogging bending.

Example nodal load input:

Node	**P**	**M**
2	-10000	0
3	0	5000
4	0	-5000

This represents a 10 000 N downward point load at node 2, a 5 000 Nm counter-clockwise nodal moment at node 3, and a 5 000 Nm clockwise nodal moment at node 4.

3 Units

AnsyBeam does not enforce a specific unit system. The user must use a consistent set of units for all input quantities.

4 Theory

AnsyBeam uses one-dimensional, two-node Euler–Bernoulli beam elements for planar bending analysis. The formulation is based on the direct stiffness method and assumes small deflections and linear elastic behaviour.

4.1 Euler–Bernoulli Beam Assumption

Euler–Bernoulli beam theory assumes that plane sections remain plane and normal to the neutral axis after deformation. Therefore, shear deformation is neglected and the transverse deflection field is governed by bending deformation only.

For a slender beam, the curvature is related to the transverse displacement by:

$$\kappa \approx \frac{d^2v}{dx^2}$$

and the bending moment is related to curvature by:

$$M = EI\kappa$$

where E is Young's modulus and I is the second moment of area.

4.2 Beam Element Degrees of Freedom

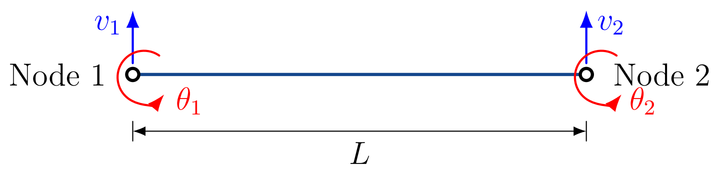


Figure 1: Two-node Euler–Bernoulli beam element with nodal degrees of freedom.

Each beam node has two degrees of freedom: the vertical displacement v and the rotation θ. As shown in Figure 1, a two-node Euler–Bernoulli beam element therefore has four element degrees of freedom.

$$\{d_e\} = \begin{Bmatrix} v_1 \\ \theta_1 \\ v_2 \\ \theta_2 \end{Bmatrix}$$

4.3 Element Stiffness Matrix

For an Euler–Bernoulli beam element with length L, Young's modulus E, and second moment of area I, the local element stiffness matrix is:

$$[k_e] = \frac{EI}{L^3} \begin{bmatrix} 12 & 6L & -12 & 6L \\ 6L & 4L^2 & -6L & 2L^2 \\ -12 & -6L & 12 & -6L \\ 6L & 2L^2 & -6L & 4L^2 \end{bmatrix}$$

The element length is calculated as:

$$L = x_j - x_i$$

where x_i and x_j are the start and end node coordinates. Elements must be defined from left to right so that:

$$L > 0$$

4.4 Equivalent Nodal Loads for UDL

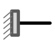
For a full-element uniformly distributed load w, the consistent equivalent nodal load vector is:

$$\{f_{UDL}\} = \begin{Bmatrix} wL/2 \\ wL^2/12 \\ wL/2 \\ -wL^2/12 \end{Bmatrix}$$

where w is positive upward and negative downward. In typical gravity loading cases, w is entered as a negative value.

4.5 Boundary Conditions

Supports are converted into restrained degrees of freedom. The available support types are:

Support Type	**Symbol**	**Fix v**	**Fix θ**	**Description**
Fixed		Yes	Yes	Vertical displacement and rotation restrained
Pinned		Yes	No	Vertical displacement restrained, rotation free
Roller		Yes	No	Vertical displacement restrained, rotation free

For this beam bending formulation, pinned and roller supports have the same analytical restraint condition:

$$v = 0, \quad \theta \text{ free}$$

They are displayed differently only for visual clarity.

5 App Panels

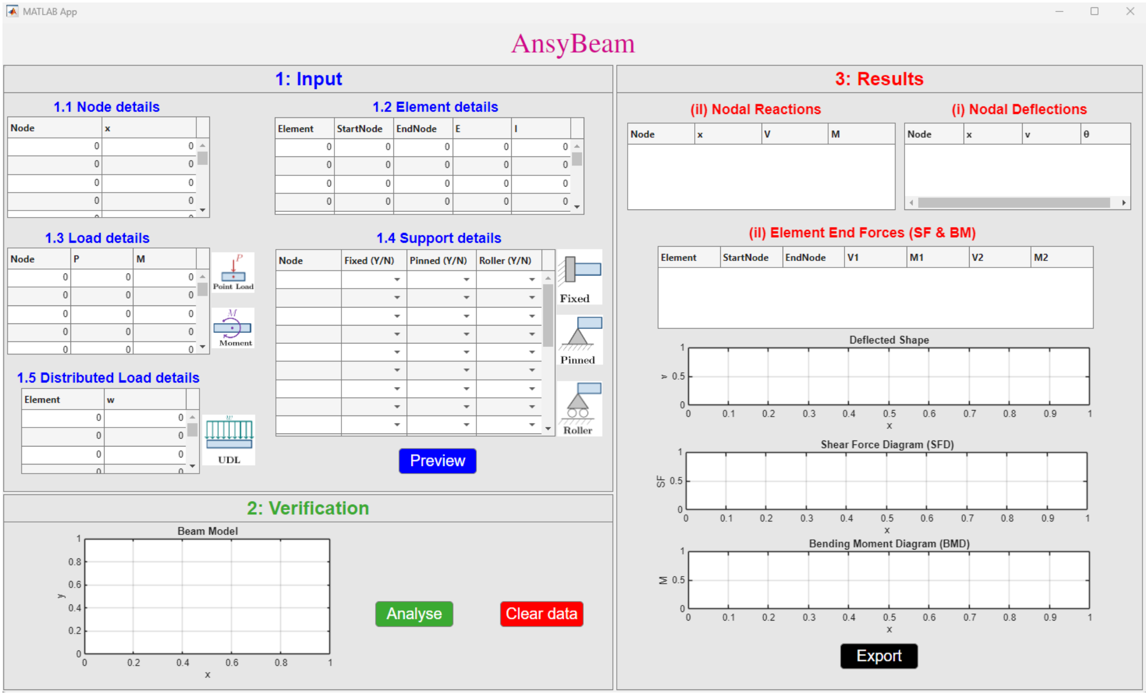


Figure 2: Overview of the AnsyBeam graphical user interface showing the main app panels.

The GUI is organised into three main panels: the **Input Panel**, the **Verification Panel**, and the **Results Panel**. These panels provide a clear workflow for defining the beam model, verifying the entered geometry, supports, and loads, and reviewing the analysis results.

5.1 Input Panel

The Input Panel is used to define the beam model. It allows the user to enter the required model data, including node coordinates, element connectivity, material and section properties, support conditions, nodal loads, applied nodal moments, and uniformly distributed loads. The five main input tables are:

- Node Data
- Element Data

- Support Data
- Load Data
- Distributed Load Data

Users can type the required values directly into the tables. Blank rows may be left unused.

5.2 Verification Panel

The Verification Panel is used to check the beam model before analysis. After clicking the **Preview** button, the app plots the beam geometry and displays the entered model visually. The verification plot displays:

- Node numbers
- Element numbers
- Fixed, pinned, and roller support symbols
- Applied point loads with directional arrows
- Applied nodal moments with curved arrows
- Uniformly distributed loads with repeated arrows

This step helps confirm that the geometry, connectivity, support conditions, point loads, nodal moments, and distributed loads have been entered correctly before running the analysis. The **Clear data** button is used to clear the entered input data from the input tables.

5.3 Results Panel

The Results Panel displays the outputs of the beam analysis. After clicking the **Analyse** button, the app shows:

- Nodal reaction table
- Nodal deflection and rotation table
- Element end force table
- Deflected shape
- Shear force diagram
- Bending moment diagram

The **Export** button is provided to generate and save the analysis report in PDF format.

6 Preloaded Example and Input Data Format

AnsyBeam opens with a preloaded example to demonstrate the required input format. The default model is a simply supported beam with a uniformly distributed load applied over the left half of the span as displayed in Figure 3.

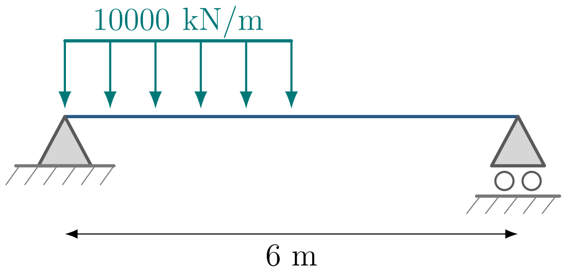


Figure 3: Simply supported beam with a UDL over the left half of the span, with moment of inertia area $I = 0.005$ m^4 and Young's modulus $E = 200$ GPa..

6.1 Fixing the Coordinates

To mathematically model the beam, a global coordinate system must be defined. In AnsyBeam, the beam axis is aligned with the global x-direction. Each node is defined using only its horizontal coordinate x.

For the default example, the left support is placed at:

$$x = 0 \text{ m}$$

and the right support is placed at:

$$x = 6 \text{ m}$$

Since the uniformly distributed load acts over the left half of the span, an additional node is placed at:

$$x = 3 \text{ m}$$

This allows the UDL to be applied over a complete beam element.

6.2 Assigning Node and Element Numbers

Before entering data, assign a unique **Node number** to every node and a unique **Element number** to every beam element. For this example, the node assignments are:

- **Node 1:** Left support
- **Node 2:** End of the UDL region at midspan
- **Node 3:** Right support

The beam is divided into two elements:

- **Element 1:** From Node 1 to Node 2
- **Element 2:** From Node 2 to Node 3

These assignments determine how the following input tables are filled out.

6.3 Node Data Format

The Node Data table defines the node number and the horizontal coordinate of each node.

Node	x
1	0.0
2	3.0
3	6.0

Columns: `Node` is the node number, and `x` is the horizontal coordinate of the node.

6.4 Element Data Format

The Element Data table defines the beam elements, their connectivity, Young's modulus, and moment of inertia or second moment of area.

Element	StartNode	EndNode	E	I
1	1	2	200×10^9	8×10^{-6}
2	2	3	200×10^9	8×10^{-6}

Columns: `Element` is the element number, `StartNode` and `EndNode` define the element connectivity, `E` is Young's modulus, and `I` is the second moment of area.

Elements must be defined from left to right:

$$x_{\text{EndNode}} > x_{\text{StartNode}}$$

6.5 Support Data Format

The Support Data table defines the support condition at each supported node.

Node	Fixed	Pinned	Roller
1	N	Y	N
3	N	N	Y

Only one support type should be selected as `Y` for each supported node.

6.6 Load Data Format

The Load Data table defines nodal point loads and applied nodal moments.

Node	P	M

Columns: `Node` is the node number, `P` is the vertical point load, and `M` is the applied nodal moment.

A downward point load is entered as a negative value. A counter-clockwise nodal moment is entered as positive, while a clockwise nodal moment is entered as negative.

For the default example, no nodal point load or nodal moment is applied.

6.7 Distributed Load Data Format

The Distributed Load Data table defines uniformly distributed loads applied over complete beam elements.

Element	**w**
1	-10000

Columns: `Element` is the element number over which the distributed load acts, and `w` is the UDL intensity.

A downward UDL is entered as a negative value. In this example, the UDL is applied only to Element 1, which represents the left half of the beam span.

6.8 Important Modelling Note

Uniformly distributed loads in AnsyBeam must act over complete elements. If a UDL acts over only part of a beam span, the beam must be split by creating nodes at the start and end of the loaded region.

7 Using the App

Step 1: Enter Input Data

Enter the node data, element data, support data, nodal load data, and distributed load data into the corresponding tables in the Input Panel, as shown in Figure 4. **Use consistent units throughout the model**. For example, if length is entered in metres and force in newtons, then Young's modulus, second moment of area, moments, and distributed loads must also be entered in compatible units.

Step 2: Preview the Beam Model

Click the **Preview** button. The app checks the input data and displays the verification plot, as shown in Figure 5. The verification stage checks for empty tables, duplicate nodes, invalid start/end nodes, non-positive element lengths, invalid material or section properties, invalid support locations, invalid load locations, and invalid distributed load assignments.

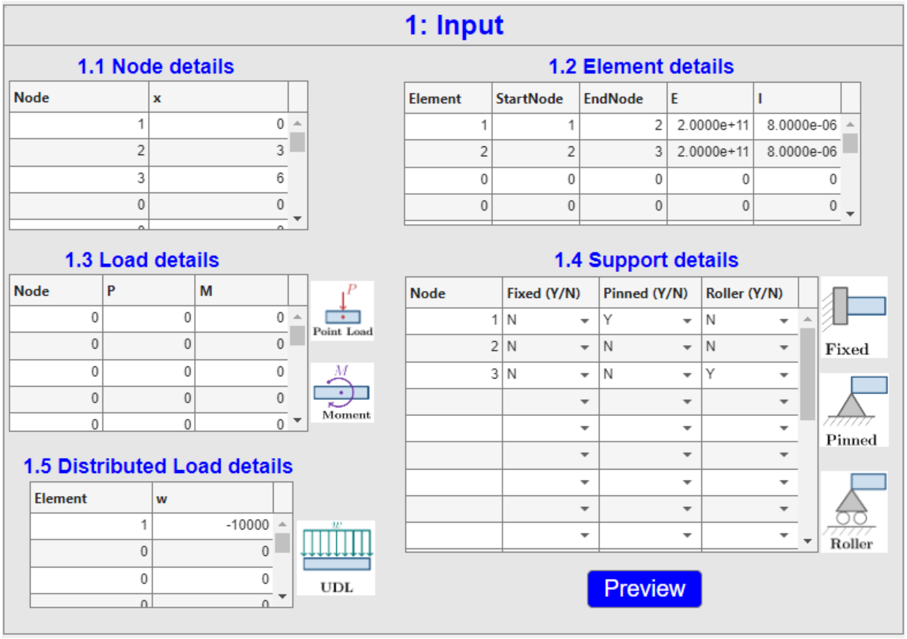


Figure 4: Input Panel populated with the data from the preloaded AnsyBeam example.

The verification plot displays the beam elements, node numbers, element numbers, support symbols, point loads, nodal moments, and uniformly distributed loads. If the displayed model does not appear correct, return to the input tables and edit the data.

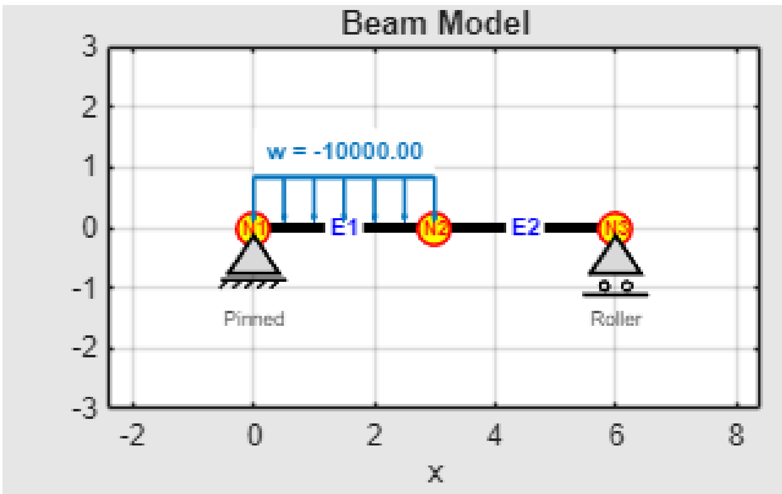


Figure 5: Verification Panel showing beam elements, node numbers, element numbers, supports, point loads, nodal moments, and distributed loads.

Step 3: Analyse the Beam

Click the **Analyse** button. The app solves the beam model using the direct stiffness method and displays the results in the Results Panel, as shown in Figure 6. The output includes nodal deflections and rotations, support reactions, element end forces, the deflected shape, shear force diagram, bending moment diagram, and summary values.

The deflected shape is recovered from the solved nodal degrees of freedom using Euler–Bernoulli beam interpolation functions. The shear force and bending moment diagrams are then reconstructed analytically from global equilibrium using the calculated support reactions and the applied loads. Positive bending moment represents sagging, while negative bending moment represents hogging.

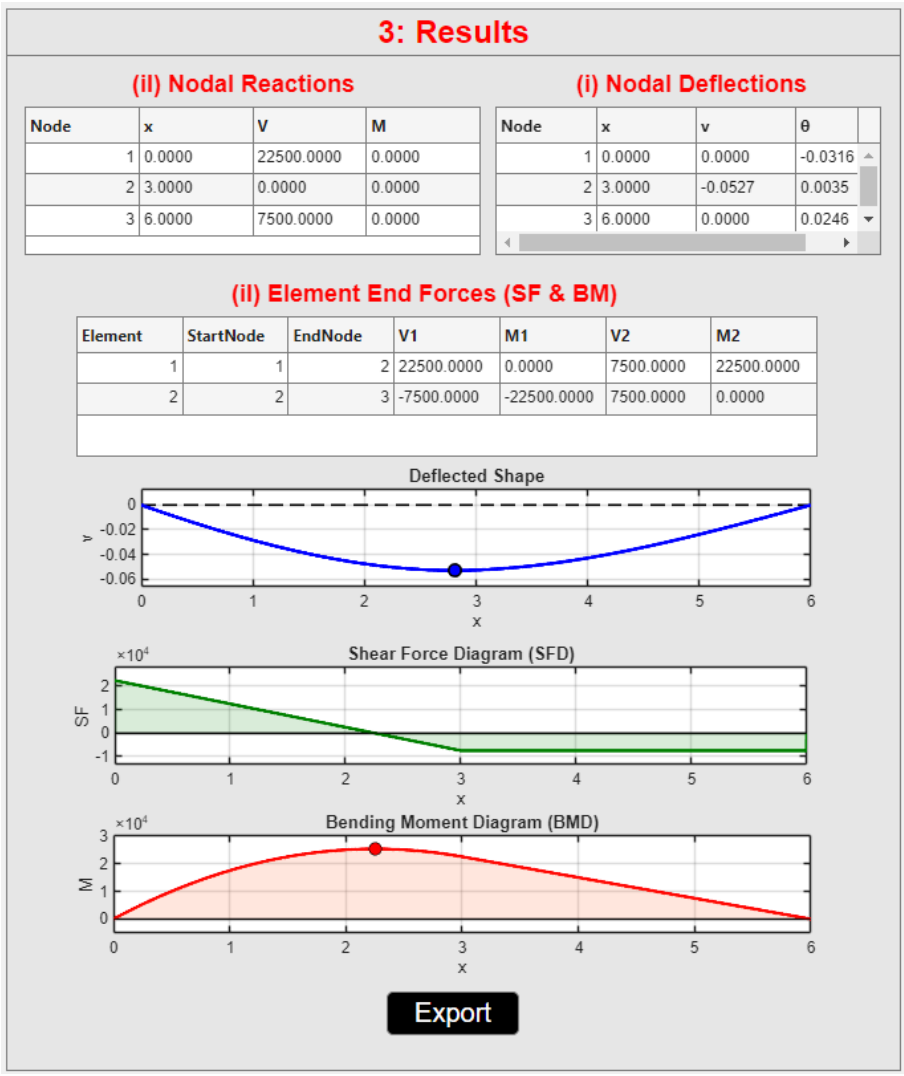


Figure 6: Results Panel displaying output tables, deflected shape, shear force diagram, and bending moment diagram.

Step 4: Export PDF Report

Click the **Export PDF Report** button to generate an A4 PDF report. The report contains the verification plot, input tables, result plots, output tables, and summary values. Wait for the pop-up message saying **"PDF report exported successfully."** before opening the file.

8 Output Results

8.1 Nodal Deflection and Rotation

The nodal deflection table reports:

Node	**x**	v	θ
1	0.0000	0.0000	0.0141
2	3.0000	0.0281	0.0000
3	6.0000	0.0000	-0.0141

where v is vertical displacement and θ is rotation.

8.2 Nodal Reactions

The nodal reaction table reports:

Node	**x**	**V**	**M**
1	0.0000	-5000.0000	0.0000
2	3.0000	0.0000	0.0000
3	6.0000	-5000.0000	0.0000

where V is the vertical reaction and M is the reaction moment.

8.3 Element End Forces

The element end force table reports the local end actions:

Element	**StartNode**	**EndNode**	V_1	M_1	V_2	M_2
1	1	2	-5000	0	5000	-15000
2	2	3	5000	15000	-5000	0

The sign of these values follows the element force vector convention used internally by the solver.

8.4 Deflected Shape

The deflected shape plot shows the transverse displacement of the beam. The undeformed beam axis is shown as a reference line.

8.5 Shear Force Diagram

The shear force diagram displays the internal shear force along the beam. Point loads appear as jumps between adjacent elements.

8.6 Bending Moment Diagram

The bending moment diagram displays the internal bending moment according to the app plotting convention. Downward/sagging-type response is shown as negative. Point moments appear as jumps in the diagram.

9 Exporting the Report

The **Export** button creates a two-page PDF report.

- Page 1 contains the verification plot and input tables.
- Page 2 contains the deflected shape, shear force diagram, bending moment diagram, output tables, and a summary box.

The summary box includes:

- Maximum deflection
- Maximum shear force
- Minimum shear force
- Maximum bending moment
- Minimum bending moment

10 Assumptions and Limitations

The current version of AnsyBeam assumes:

- Linear elastic material behaviour.
- Small displacement behaviour.
- Euler–Bernoulli beam theory.
- No axial deformation.
- No shear deformation.
- Point loads and point moments are applied only at nodes.

- UDLs act over complete elements.
- Elements are straight and aligned with the global x-axis.

and does not include:

- Timoshenko beam elements.
- Internal hinges.
- Elastic supports.
- Spring foundations.
- Moving loads.
- Linearly varying distributed loads.
- Material or geometric nonlinearities.

11 Troubleshooting

Problem	**Possible Solution**
Verification fails	Check that node numbers are unique and that all element start and end nodes exist.
Analysis does not run	Check supports, loads, connectivity, and ensure that the beam is stable and sufficiently restrained.
No load, moment, or UDL appears	Check that the load node or UDL element exists and that the load value is not zero.
Unexpected deflection values	Check that consistent units are used for E, I, length, force, moment, and distributed load.
Unexpected SFD or BMD values	Check the load signs, support locations, and UDL assignments.
PDF does not open correctly	Wait for the export-success pop-up before opening the file.

12 Summary

AnsyBeam provides a simple and structured workflow for defining, verifying, analysing, visualising, and reporting planar beam structures. As a MATLAB-based GUI tool, it supports linear elastic static analysis of Euler–Bernoulli beam models using the direct stiffness method. Developed at Auckland University of Technology (AUT), AnsyBeam is intended as an educational and engineering calculation tool for understanding beam behaviour.